Protocols & Computer Ports Architecture Notes

Protocols: What are Web Protocols?

A web protocol is simply a set of rules that define how data is sent and received over a network (especially the internet).

- **Humans** → language (English, Hindi)
- **Computers** → protocols (HTTP, HTTPS, DNS, etc.)
- *Without protocols, devices wouldn't understand each other.*

Understanding Web Protocols: How the Internet Works

- **HTTP & HTTPS:** HTTP (GET/index.html), Port 80 - Unencrypted. HTTPS, Port 443 - Encrypted, Secure Connection (SSL/TLS).
- **DNS (Domain Name System):** Translates Domain to IP Address (e.g., example.com → 192.168.1.1) via DNS Lookup.
- **SMTP (Email Protocol):** Used for sending email via Port 25/587 from You → Mail Server → Recipient.
- **FTP/SFTP (File Transfer):** FTP runs on Port 21, SFTP runs on Port 22 (Secure).
- **RDP (Remote Desktop):** Remote Desktop Control on Port 3389.

The Internet Journey:

1. DNS Lookup
2. TCP Connection
3. Data Transfer

Key Protocols & Ports Summary:

- HTTP/HTTPS: 80 / 443
- DNS: 53
- FTP/SFTP: 21 / 22
- SMTP: 25 / 587
- SSH: 22
- RDP: 3389

Where Do Web Protocols Fit?

Web protocols operate across layers of the OSI Model:

- **Application Layer:** HTTP, HTTPS, DNS, FTP
- **Transport Layer:** TCP, UDP

- **Network Layer:** IP
- **Data Link Layer:** Ethernet, WiFi

Common Protocols Deep Dive

1. HTTP - Hypertext Transfer Protocol

HTTP is the core web protocol used to transfer web pages.

- **Features:** Works on Port 80, Stateless (IMPORTANT concept).

How HTTP Works (Step-by-Step Process):

1. User issues URL from a browser (http://host:port/path/file).
2. Browser sends a request message (GET URL HTTP/1.1, Host: host:port).
3. Server maps the URL to a file or program under the document directory.
4. Server returns a response message (HTTP/1.1 200 OK) over TCP/IP.
5. Browser formats the response and displays it.

How Browser Sends Requests (In-Depth):

- Browser looks for the URL in the Browser Cache.
- If found, it goes straight to a TCP Connection.
- If not found, it initiates a request to the DNS Server first, then establishes a TCP Connection with the Web Server, which handles the request and sends back an HTTP Response.

HTTP Methods & Purpose:

- **GET:** Fetch data
- **POST:** Send data
- **PUT:** Update resource
- **DELETE:** Remove resource
- **PATCH:** Partial update

- **Stateless Nature:** HTTP does NOT remember who you are or previous requests. That's why we use Cookies, Sessions, and Tokens.

> **SOC Indicators:** Suspicious URLs, Repeated requests (/login brute force), Unusual user-agents, Large POST requests.
> **Attacks:** XSS, SQL Injection, Credential harvesting, HTTP Flood (DDoS).

HTTP Status Codes

HTTP response status codes indicate whether a specific HTTP request has been successfully completed. Responses are grouped in five classes:

1. Informational responses (100 - 199)
2. Successful responses (200 - 299)

3. Redirection messages (300 - 399)
4. Client error responses (400 - 499)
5. Server error responses (500 - 559)

- **200:** OK
- **301:** Redirect
- **400:** Bad request
- **401:** Unauthorized
- **403:** Forbidden
- **404:** Not found
- **500:** Server error

2. HTTPS (Secure HTTP)

- **What is it?** HTTP + Encryption (via SSL/TLS) running on Port 443.
- **Provides:** Confidentiality, Integrity, and Authentication.
- **Why it matters:** Without HTTPS, MITM (Man-In-The-Middle) attacks are possible, credentials can be sniffed, and data is readable in plaintext.

3. DNS (Domain Name System)

DNS is the phonebook of the Internet. Humans access information online through domain names (like google.com), while web browsers interact through IP addresses. DNS translates domain names to IP addresses.

- **Specs:** Works on Port 53, Uses UDP (mostly).
- **DNS Resolution Process Hierarchy:** Browser cache → OS cache → Resolver (ISP) → Root server → TLD server → Authoritative server.
- **Query Types (Iterative vs. Recursive):**
  - *Iterative Query:* The requesting host contacts the local DNS server, which iteratively polls the Root DNS server, TLD DNS server, and Authoritative DNS server on behalf of or returning pointers to the client.
  - *Recursive Query:* The burden of resolution passes up the chain step-by-step to the Authoritative server and bubbles straight back down.

> **SOC Indicators:** Too many DNS requests, Suspicious domains, Random subdomains, DNS tunneling.
> **Attacks:** DNS Spoofing, DNS Tunneling (data exfiltration), DGA (Domain Generation Algorithm).

4. FTP / SFTP (File Transfers)

Used to transfer files between systems.

- **FTP:** Port 21, Plaintext (NOT secure), Modes: Active and Passive.
- **SFTP:** Runs over SSH on Port 22, Encrypted.

> **SOC Indicators:** Large file transfers, Unknown uploads, Anonymous login.
> **Attacks:** Credential sniffing (FTP), Data exfiltration, Brute force login.

5. SMTP (Email Sending)

- **What it does:** Sends emails across the internet from Sender → SMTP Mail Server → Receiver (via POP3/ protocols).
- **Port:** 25 (SMTP), 587 (secure submission).

> **SOC Indicators:** Bulk email sending, Unknown sender domains, Email spoofing.
> **Attacks:** Phishing, Spam campaigns, Email spoofing.

6. SSH (Secure Shell)

- **What it does:** Secure remote login on Port 22.

> **SOC Indicators:** Multiple failed logins, Login from unusual IP, Root login attempts.
> **Attacks:** Brute force, Credential theft, Unauthorized access.

7. RDP (Remote Desktop Protocol)

- **What it does:** Remote control of Windows systems on Port 3389.

> **SOC Indicators:** Login at odd hours, Multiple login attempts, External IP login.
> **Attacks:** RDP brute force, Lateral movement, Ransomware entry point.

8. SMB (Server Message Block)

- **What it does:** File sharing in Windows networks on Port 445.

> **SOC Indicators:** Access to admin shares (C$), File movement across systems, Lateral movement.
> **Attacks:** EternalBlue exploit, SMB relay, Lateral movement.

Computer Ports Architecture

What is a Computer Port?
A port is a virtual point where network connections start and end. Ports are software-based and managed by a computer's operating system. Each port is associated with a specific process or service. They allow computers to easily differentiate between traffic types (e.g., separating emails from webpages on the same connection).

There are **65,535** possible port numbers divided into groups:

- **Well-known Ports (0 - 1023):** Reserved for standard services like FTP (21), SSH (22), SMTP (25), DNS (53), HTTP (80), HTTPS (443).
- **Registered Ports (1024 - 49151):** Used by specific applications like databases (MySQL 3306, SQL Server 1433).
- **Dynamic / Private Ports (49152 - 65535):** Temporarily assigned for client connections (e.g., outbound browser sessions).
- **Internal Ports:** Used strictly inside your private network (LAN) for local communication.

- **External Ports:** Open to the internet; routers map them to internal ports to securely route incoming traffic.

Important Ports Reference Table

Port	Protocol	Use
21	FTP	File transfer
22	SSH / SFTP	Secure remote access / file transfer
25	SMTP	Email sending
53	DNS	Domain resolution
80	HTTP	Web traffic
443	HTTPS	Secure web traffic
3389	RDP	Remote desktop
445	SMB	File sharing

Real SOC Log Example

- **Source IP:** 185.221.x.x
- **Destination IP:** 192.168.1.10
- **Port:** 3389
- **Attempts:** 500
- **Status:** Failed

SOC Analyst Analysis: Immediately think **RDP brute force attack**.